

3 — Featurization: filtrations, persistence images, summary functions

1 Diagrams

Compute and vectorize H_0 and H_1 persistence diagrams of each cloud. We consider multiple filtrations: Rips, Alpha, and DTM_k . The latter is swept over $k \in \{5, 10, 15, 20\}$.

2 Persistence images

We follow (Adams et al. 2017), with some small adjustments.

Procedure

- **Rescale.** Multiply all coordinates by $\sqrt{n}$: lengths are then in units of the mean point spacing $1/\sqrt{n}$, and the image is divided by n . Without this a pattern at $\bar{n} = 800$ occupies a third of the axis of one at $\bar{n} = 100$ and carries $\approx 7\times$ the mass, so resolution and dynamic range would both track $\bar{n}$
- **Coordinate transformation.** (birth, death) $\rightarrow$ (birth, persistence)
- **Smooth.** Each pair becomes a Gaussian of width σ centred on it, weighted linearly by its persistence (diagonal points are weightless)
- **Rasterise.** Pixel values are the integral of the surface over its box
- **2-D or 1-D.** H_1 is always 2-D. H_0 is 2-D only under DTM; Rips and alpha have degenerate H_0 births ($= 0$), so their H_0 image is the persistence axis alone
- **Standardise.** Images pass through $\sqrt{\cdot}$, then a z -score with one (mean, sd) per filtration channel. A linearly weighted H_1 image is $\approx 93\%$ empty with a few very bright pixels. $\sqrt{\cdot}$ is variance-stabilising.
- Resolution: 64×64
- $\sigma = 1$ pixel

Calibration

The axes of the persistence diagrams are calibrated. Diagrams across the same homology dimension need to share the same axes in order for the vectorization to be meaningful.

- Fitted on training clouds only, once per (filtration, dimension), then frozen and applied to validation and test. The z -score is fitted the same way
- Each training diagram contributes its own min birth, max birth and max persistence, so clipping is a property of a whole pattern rather than of its busiest corner
- The box holds the full support of **coverage** = 99% of training diagrams:

$$\text{persistence} \in [0, 1.05 \cdot q_{99}(\text{max persistence})], \quad \text{birth} \in [q_1(\text{min birth}), 1.05 \cdot q_{99}(\text{max birth})],$$

the factor 1.05 a pad so the 99% diagram's own tail is not on the edge

3 Summary functions

The classical arm featurizes the same clouds without a filtration.

Procedure

Let $\hat{\phi}$ be any estimator taking a cloud x to a real function of a single scale, $r \mapsto \hat{\phi}_x(r)$ on $(0, r_{\max}]$. Featurizing it is sampling it at 512 radii.

- **Sample.** Fix a grid $0 < r_1 < \dots < r_K = r_{\max}$ shared by every cloud, and take

$$x \mapsto (\hat{\phi}_x(r_1), \dots, \hat{\phi}_x(r_{512})) \in \mathbb{R}^{512}$$

- **Standardise.** A z -score per function: one (mean, sd) pooled over training clouds *and* radii, so the curve is shifted and rescaled, never reshaped per radius or per cloud
- No need for $\sqrt{\cdot}$: bounded functions, not a sparse measure with a heavy tail

Nothing here is specific to the four summary functions below. Any estimator indexed by a scale (e.g. pair correlation function, Betti curve, $\hat{K}$ under a different edge correction) vectorizes the same way. The only choices are $r_{\max}$ and where the K samples sit.

Functions

- $L(r) - r = \sqrt{\hat{K}_x(r)/\pi} - r$ (isotropic edge correction), $= 0$ under CSR
- $F(r)$: empty-space function, the distance from a fixed location to the nearest point. Sees voids
- $G(r)$: nearest-neighbour function, the distance from a typical point to its nearest other point. Sees short-range clumping or inhibition that $\hat{K}$ averages over
- $J(r) = (1 - G)/(1 - F)$: $= 1$ under CSR, < 1 clustering, > 1 inhibition

Estimation details:

- F and G use the reduced-sample (border) correction: keep only the points, or test locations, further than r from the edge of W , so nothing is censored by the window. Exactly unbiased, no tuning constant
- F 's test locations are a 64×64 interior grid, $\approx 15\times$ oversampled against the sweep's median $n = 282$
- J is held flat past the last radius with $1 - F > 0.05$ and clipped at 5: an unbounded ratio entering a z -scored channel would swamp it

Grid

Two conventions, because the four functions do not vary on the same axis.

- **fixed:** $r \in (0, 0.25]$ in 512 equal steps
- **sqrtn:** $r = u/\sqrt{n}$, $u \in (0, 2]$ (same $\sqrt{n}$ rescaling the images apply to diagram coordinates)

F and G are distance CDFs and reach 1 near the mean nearest-neighbour distance $\approx 1/(2\sqrt{n})$. On **fixed** they sit at exactly 1 over 62% and 66% of their samples, against 23% and 31% on **sqrtn**, and the saturation point moves by $4.7\times$ across the sweep's range of n (44–961): one encoder then sees the same rise at very different positions.